

MAT1503

October/November 2015

LINEAR ALGEBRA

Duration 2 Hours

100 Marks

EXAMINERS

FIRST

SECOND

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Closed book examination

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This paper consists of 3 pages

Answer All Questions

[TURN OVER]

QUESTION 1

- (a) Express the following matrix equation

$$x_1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ -4 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

into a system of linear equations and show that $x_1 = 2, x_2 = 3, x_3 = 4$ is a solution to the matrix equation and to the resulting system of linear equations (6)

- (b) Find
- 1×3
- matrices
- X, Y
- such that

$$\begin{aligned} X + 2Y &= \begin{bmatrix} 1 & 3 & -2 \end{bmatrix} \\ X + Y &= \begin{bmatrix} 2 & 0 & 1 \end{bmatrix} \end{aligned} \quad (6)$$

- (c) Let

$$A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$(i) \text{ compute } A^{-1} \quad (3)$$

$$(ii) \text{ using } A^{-1} \text{ computed in (i) above, find a } 2 \times 2 \text{ matrix } L \text{ such that } AL = B \quad (3)$$

$$(iii) \text{ using } A^{-1} \text{ computed in (i) above, find a } 2 \times 2 \text{ matrix } M \text{ such that } MA = B \quad (3)$$

- (d) Let
- C
- be a matrix and
- k
- be a scalar such that
- $kC = 0$
- . Then show that either
- $k = 0$
- or
- $C = 0$

(4)

[25]

QUESTION 2

- (a) Let
- D
- and
- E
- be
- $n \times n$
- matrices

$$(i) \text{ if } D \text{ is invertible, then show that } D^{-1} \text{ is also invertible and that } (D^{-1})^{-1} = D \quad (5)$$

$$(ii) \text{ show that } DE \text{ is invertible if and only if } D \text{ and } E \text{ are both invertible} \quad (5)$$

- (b) Using Cramer's Rule, solve the following system of linear equations

$$\begin{cases} 2x_1 + x_2 - 3x_3 = 0 \\ 4x_1 + 5x_2 + x_3 = 8 \\ -2x_1 - x_2 + 4x_3 = 2 \end{cases}$$

(7)

[TURN OVER]

(c) Given the following matrix

$$F = \begin{bmatrix} a & 0 & 0 & 0 \\ u & b & 0 & 0 \\ v & w & c & 0 \\ x & y & z & d \end{bmatrix}$$

then show that $\det(F)$ is the product of the diagonal entries (4)

(d) For what values of k does the matrix

$$G = \begin{bmatrix} 1 & 0 & -k \\ -1 & 3 & 1 \\ 0 & 2k & -4 \end{bmatrix}$$

have an inverse (4)

[25]

QUESTION 3

Consider the vectors $\underline{u} = (1, 0, \sqrt{2})$ and $\underline{v} = (1, \sqrt{2}, 0)$ in standard position

(a) Determine the orthogonal complement of $\underline{v}$ on $\underline{u}$ (5)

(b) Calculate the area of the parallelogram determined by $\underline{u}$ and $\underline{v}$ (5)

(c) Find an equation of the plane containing $\underline{u}$ and $\underline{v}$ (5)

(d) Determine the parametric equations of the plane containing $\underline{u}$ and $\underline{v}$ (10)

[25]

QUESTION 4

(a) Use de Moivre's Theorem to express $\sin 4\theta$ in terms of powers of $\sin \theta$ and $\cos \theta$ (8)

(b) Determine the 4th roots of 16 in the form $a + ib$, where $a, b \in \mathbb{R}$ (10)

(c) Use de Moivre's Theorem to determine $(-1 + i)^{128}$ in the form $x + iy$, where $x, y \in \mathbb{R}$ (7)

[25]

TOTAL: 100 Marks